

Mathematics for Data Science

Solution Sheet – Lecture 2

Matrices of a linear application and change of basis

M1 – AI / Data Science

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Unless stated otherwise, all vector spaces are real vector spaces.

Part 1 – Course questions

Part 1(a) – Short exercises

Exercise 1

Find the matrix corresponding to the linear transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

defined by

$$T(e_1) = (1, 0, 0)^T, \quad T(e_2) = (0, 1, 0)^T, \quad T(e_3) = (0, 0, 1)^T$$

in the standard basis

$$\{e_1, e_2, e_3\}.$$

Solution. The columns of the matrix representation of a linear map in the standard basis are the images of the standard basis vectors.

Thus

$$[T] = \begin{pmatrix} | & | & | \\ T(e_1) & T(e_2) & T(e_3) \\ | & | & | \end{pmatrix}.$$

Since

$$T(e_1) = e_1, \quad T(e_2) = e_2, \quad T(e_3) = e_3,$$

we obtain

$$[T] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

Hence T is the identity transformation on $\mathbb{R}^3$.

□

Exercise 2

Solution.

We consider the linear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(x_1, x_2) = (x_1, 2x_2).$$

1. Matrix representation in the standard basis. Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

be the standard basis of $\mathbb{R}^2$.

We compute

$$T(e_1) = T(1, 0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad T(e_2) = T(0, 1) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

The columns of the matrix of T in the standard basis are therefore $T(e_1)$ and $T(e_2)$. Hence

$$[T]_{\text{std}} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

2. Matrix representation in the basis $B' = (v_1, v_2)$. with

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

To obtain the matrix of T in the basis B' , we compute $T(v_1)$ and $T(v_2)$ and express each resulting vector as a linear combination of v_1 and v_2 .

First,

$$T(v_1) = T(1, 1) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

We look for scalars $a, b \in \mathbb{R}$ such that

$$T(v_1) = av_1 + bv_2.$$

Thus,

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a+b \\ a-b \end{pmatrix}.$$

Therefore a and b satisfy

$$\begin{cases} a+b=1, \\ a-b=2. \end{cases}$$

Adding the two equations gives

$$2a = 3, \quad a = \frac{3}{2},$$

and consequently

$$b = 1 - a = -\frac{1}{2}.$$

Hence

$$T(v_1) = \frac{3}{2}v_1 - \frac{1}{2}v_2,$$

so that

$$[T(v_1)]_{B'} = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Similarly,

$$T(v_2) = T(1, -1) = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

We look for $a, b \in \mathbb{R}$ such that

$$T(v_2) = av_1 + bv_2.$$

This gives

$$\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} a+b \\ a-b \end{pmatrix},$$

and hence

$$\begin{cases} a+b=1, \\ a-b=-2. \end{cases}$$

Adding the two equations gives

$$2a = -1, \quad a = -\frac{1}{2},$$

and therefore

$$b = 1 - a = \frac{3}{2}.$$

Thus

$$T(v_2) = -\frac{1}{2}v_1 + \frac{3}{2}v_2,$$

so that

$$[T(v_2)]_{B'} = \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix}.$$

The columns of the matrix representation of T in the basis B' are the B' -coordinates of $T(v_1)$ and $T(v_2)$. Therefore,

$$[T]_{B'} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

□

NB: the matrix representation of a linear transformation *depends in general* on the chosen basis. But a change of basis does not necessarily change the matrix: for example, the identity transformation has matrix I_n in every basis.

Remark: Another way of looking at this exercise.

The matrix whose columns are the new basis vectors expressed in the standard basis is

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It satisfies

$$[x]_B = P[x]_{B'}.$$

For an endomorphism, the matrix in the new basis is

$$[T]_{B'} = P^{-1}AP.$$

Where

$$A = [T]_B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Remarking that

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

we obtain

$$[T]_{B'} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Therefore, again, we obtain

$$[T]_{B'} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

Exercise 3

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Verify directly that the (i, j) entry of AB is

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Solution. Let

$$A = (a_{ik})_{1 \leq i \leq m, 1 \leq k \leq n}$$

and

$$B = (b_{kj})_{1 \leq k \leq n, 1 \leq j \leq p}.$$

The j -th column of B is

$$B_{\cdot j} = \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}.$$

The j -th column of AB is therefore

$$AB_{\cdot j}.$$

Its i -th component is the scalar product of the i -th row of A with the j -th column of B :

$$(AB)_{ij} = (a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}) \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}.$$

Hence

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

□

Exercise 4

Compute the rank and kernel of the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Determine $\text{rank}(A)$ and $\dim(\ker(A))$.

Solution. The matrix defines a linear map

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

given by

$$A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Method 1: the family of columns of A is not linearly independent. But taking the first two column vectors, it is.

Method 2: (with point of view from Gauss pivot): The two rows of A are linearly independent, and there are two pivot columns.

Hence

$$\boxed{\text{rank}(A) = 2}.$$

The rank-nullity theorem already gives us that the kernel of A has dimension 1. But we want to describe it more precisely.

To find the kernel, we solve

$$Ax = 0.$$

This gives

$$x_1 = 0, \quad x_2 = 0,$$

while x_3 is arbitrary.

Therefore

$$\ker(A) = \left\{ \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix} : t \in \mathbb{R} \right\} = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Consequently,

$$\boxed{\dim(\ker(A)) = 1}.$$

Again, this is consistent with the rank-nullity theorem:

$$\text{rank}(A) + \dim(\ker(A)) = 2 + 1 = 3.$$

□

Exercise 5

Are the following matrices invertible? If so, compute their inverse.

1.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

2.

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution.

1. The matrix A is diagonal and both diagonal entries are nonzero. Therefore A is invertible.

Its inverse is obtained by taking the reciprocal of each diagonal entry:

$$\boxed{A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}}.$$

Indeed,

$$AA^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = I_2.$$

2. The matrix B is not invertible. For example,

$$B \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

so

$$\ker(B) \neq \{0\}.$$

Equivalently,

$$\det(B) = 0.$$

Hence

$$\boxed{B \text{ is not invertible.}}$$

3. $C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. To compute its inverse, we apply Gaussian elimination to the augmented matrix

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right).$$

We first eliminate the entry above the pivot in the third column:

$$R_2 \leftarrow R_2 - R_3.$$

This gives

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right).$$

We then eliminate the entry above the pivot in the second column:

$$R_1 \leftarrow R_1 - R_2.$$

Hence,

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right).$$

The left-hand side is now the identity matrix, so the right-hand side is C^{-1} . Therefore,

$$\boxed{A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}.$$

We can verify the result by computing

$$CC^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

4. $D = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

applying Gaussian elimination to

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right).$$

First, perform

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - R_1.$$

We obtain

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right).$$

Next, multiply the second row by -1 :

$$R_2 \leftarrow -R_2,$$

which gives

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right).$$

We eliminate the entry in the third column of the second row:

$$R_2 \leftarrow R_2 + R_3.$$

Thus,

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right).$$

Finally, eliminate the entry above the pivot in the second column:

$$R_1 \leftarrow R_1 - R_2.$$

We obtain

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right).$$

Therefore,

$$\boxed{D^{-1} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{pmatrix}}.$$

□

Exercise 6

Let $A \in \mathbb{R}^{n \times n}$. Prove that the following statements are equivalent:

1. The transformation $x \mapsto Ax$ is bijective.
2. $\text{Im}(A) = \mathbb{R}^n$.
3. $\ker(A) = \{0\}$.

Solution. Let

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad T(x) = Ax.$$

We use the rank–nullity theorem:

$$\dim(\ker A) + \operatorname{rank}(A) = n.$$

(1) $\Rightarrow$ (2).

If T is bijective, then in particular T is surjective. Therefore every vector in $\mathbb{R}^n$ is the image of some vector in $\mathbb{R}^n$, so

$$\operatorname{Im}(A) = \mathbb{R}^n.$$

(2) $\Rightarrow$ (3).

Suppose

$$\operatorname{Im}(A) = \mathbb{R}^n.$$

Then

$$\operatorname{rank}(A) = \dim(\operatorname{Im}(A)) = n.$$

By the rank–nullity theorem,

$$\dim(\ker A) = n - \operatorname{rank}(A) = 0.$$

Hence

$$\ker(A) = \{0\}.$$

(3) $\Rightarrow$ (1).

Suppose

$$\ker(A) = \{0\}.$$

Then

$$\dim(\ker A) = 0.$$

Again by rank–nullity,

$$\operatorname{rank}(A) = n.$$

Thus

$$\dim(\operatorname{Im}(A)) = n.$$

Since $\operatorname{Im}(A)$ is an n -dimensional subspace of $\mathbb{R}^n$, it follows that

$$\operatorname{Im}(A) = \mathbb{R}^n.$$

Hence T is surjective.

Moreover, since

$$\ker(T) = \{0\},$$

T is injective.

Therefore T is bijective.

We have proved

$$\boxed{T \text{ bijective} \iff \operatorname{Im}(A) = \mathbb{R}^n \iff \ker(A) = \{0\}}.$$

□

Exercise 7 (*)

Find some examples of 3×3 orthogonal matrices.

Solution. Recall that a matrix $Q \in \mathbb{R}^{3 \times 3}$ is orthogonal if

$$Q^T Q = I_3.$$

Equivalently, its columns form an orthonormal basis of $\mathbb{R}^3$. Several examples are given below.

The identity matrix:

$$Q_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A reflection:

$$Q_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

A permutation matrix:

$$Q_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A rotation of angle θ around the third coordinate axis:

$$Q_4 = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Indeed,

$$Q_4^T Q_4 = I_3.$$

Notice that orthogonal matrices may have determinant 1 or -1 : rotations have determinant 1, while reflections may have determinant -1 . $\square$

Part 1(b) – True/False questions

For each statement, decide whether it is true or false, and justify your answer.

T/F 1. A matrix $A \in \mathbb{R}^{m \times n}$ is invertible if and only if $m = n$.

Solution. False.

Being square is necessary for a matrix to have an ordinary two-sided inverse, but it is not sufficient.

For example,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

is square but not invertible.

For a square matrix $A \in \mathbb{R}^{n \times n}$,

$$A \text{ is invertible} \iff \text{rank}(A) = n \iff \ker(A) = \{0\}.$$

$\square$

T/F 2. If A is a diagonal matrix, then A^2 is also diagonal.

Solution. True.

If

$$A = \text{diag}(\lambda_1, \dots, \lambda_n),$$

then

$$A^2 = \text{diag}(\lambda_1^2, \dots, \lambda_n^2).$$

Therefore A^2 is diagonal. □

T/F 3. If A^2 is diagonal, then A is also diagonal.

Solution. False.

Consider

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is not diagonal, but

$$A^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

which is diagonal. □

T/F 4. If a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is injective, then it is also surjective.

Solution. True.

Since T is injective,

$$\ker(T) = \{0\}.$$

Thus

$$\dim(\ker T) = 0.$$

By rank-nullity,

$$\text{rank}(T) + \dim(\ker T) = n,$$

so

$$\text{rank}(T) = n.$$

Hence

$$\dim(\text{Im}(T)) = n.$$

Since $\text{Im}(T) \subseteq \mathbb{R}^n$, this implies

$$\text{Im}(T) = \mathbb{R}^n.$$

Thus T is surjective.

This result relies on the fact that the domain and codomain have the same finite dimension.

□

T/F 5. If a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is surjective, then its matrix representation in the standard basis is invertible.

Solution. True.

If T is surjective, then

$$\text{Im}(T) = \mathbb{R}^n,$$

so

$$\text{rank}(T) = n.$$

The matrix $A \in \mathbb{R}^{n \times n}$ representing T therefore has full rank:

$$\text{rank}(A) = n.$$

A square matrix of full rank is invertible. Hence A is invertible.

□

Part 2 – Computational exercises

Exercise 1 (Matrix representation of a linear transformation). Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be the linear transformation defined by

$$T(x_1, x_2, x_3) = (x_1 + 2x_2, x_2 - x_3, 3x_1 + x_3, 2x_1 - x_2 + x_3).$$

Solution.

1. Matrix representation in the standard bases The columns of the matrix representation are the images of the standard basis vectors.

We have

$$T(e_1) = T(1, 0, 0) = \begin{pmatrix} 1 \\ 0 \\ 3 \\ 2 \end{pmatrix},$$

$$T(e_2) = T(0, 1, 0) = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -1 \end{pmatrix},$$

and

$$T(e_3) = T(0, 0, 1) = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 1 \end{pmatrix}.$$

Therefore,

$$[T]_{C \leftarrow B} = A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix}.$$

Indeed, for every $x \in \mathbb{R}^3$,

$$[T(x)]_C = A[x]_B.$$

NB: other way seeing this

$$A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ x_2 - x_3 \\ 3x_1 + x_3 \\ 2x_1 - x_2 + x_3 \end{pmatrix}.$$

2. Unicity of the matrix representation Suppose A and A' are two matrices representing T in the same standard bases.

Then, for every standard basis vector e_j ,

$$Ae_j = T(e_j) = A'e_j.$$

But Ae_j is precisely the j -th column of A , and $A'e_j$ is the j -th column of A' . Thus every column of A equals the corresponding column of A' . Therefore

$$A = A'.$$

Hence the matrix representation of T in fixed bases is unique.

3. Rank and kernel We determine the rank using Gaussian elimination. Starting from

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix},$$

perform

$$R_3 \leftarrow R_3 - 3R_1, \quad R_4 \leftarrow R_4 - 2R_1.$$

This gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -6 & 1 \\ 0 & -5 & 1 \end{pmatrix}.$$

Next, perform

$$R_1 \leftarrow R_1 - 2R_2, \quad R_3 \leftarrow R_3 + 6R_2, \quad R_4 \leftarrow R_4 + 5R_2.$$

We obtain

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \\ 0 & 0 & -4 \end{pmatrix}.$$

Finally,

$$R_4 \leftarrow 5R_4 - 4R_3,$$

which gives

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & -5 \\ 0 & 0 & 0 \end{pmatrix}.$$

There are three pivot rows. Hence

$$\boxed{\text{rank}(A) = 3}.$$

Since there is a pivot in every column, the homogeneous system

$$Ax = 0$$

has only the trivial solution. Therefore,

$$\ker(A) = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

Equivalently,

$$\ker(T) = \{0\}.$$

4. Injectivity, surjectivity, and bijectivity Since

$$\ker(T) = \{0\},$$

the transformation T is injective.

On the other hand,

$$\text{rank}(T) = 3,$$

whereas the codomain has dimension

$$\dim(\mathbb{R}^4) = 4.$$

Thus,

$$\text{range}(T) \neq \mathbb{R}^4,$$

so T is not surjective.

Consequently,

$$T \text{ is injective, but not surjective and therefore not bijective.}$$

5. Matrix representation in the basis B' Let

$$B' = (v_1, v_2, v_3),$$

where

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

We assume that the basis C of the codomain remains the standard basis of $\mathbb{R}^4$.

The columns of $[T]_{C \leftarrow B'}$ are the vectors $T(v_1)$, $T(v_2)$, and $T(v_3)$ expressed in the basis C .

First,

$$T(v_1) = T(1, 1, 0) = \begin{pmatrix} 3 \\ 1 \\ 3 \\ 1 \end{pmatrix}.$$

Similarly,

$$T(v_2) = T(0, 1, 1) = \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

and

$$T(v_3) = T(1, 0, 1) = \begin{pmatrix} 1 \\ -1 \\ 4 \\ 3 \end{pmatrix}.$$

Therefore,

$$[T]_{C \leftarrow B'} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & -1 \\ 3 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix}.$$

5. Change of basis NB: Carefully define the notations you use for change-of-basis matrices. Here we use the following notations: the columns of

$$P_{B \leftarrow B'}$$

are the vectors of B' expressed in the standard basis B . Therefore,

$$P_{B \leftarrow B'} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

This matrix satisfies

$$[x]_B = P_{B \leftarrow B'} [x]_{B'}.$$

We now find the change-of-coordinates matrix in the opposite direction,

$$P_{B' \leftarrow B}.$$

Let

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

Suppose

$$[x]_{B'} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Then

$$x = av_1 + bv_2 + cv_3.$$

Thus,

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + c \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Comparing coordinates gives

$$\begin{cases} a + c = x_1, \\ a + b = x_2, \\ b + c = x_3. \end{cases}$$

Adding the first two equations and subtracting the third gives

$$2a = x_1 + x_2 - x_3,$$

so

$$a = \frac{x_1 + x_2 - x_3}{2}.$$

Similarly,

$$b = \frac{-x_1 + x_2 + x_3}{2},$$

and

$$c = \frac{x_1 - x_2 + x_3}{2}.$$

Therefore,

$$[x]_{B'} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} [x]_B.$$

Hence

$$P_{B' \leftarrow B} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Notice that we have obtained this matrix by explicitly solving the coordinate equations. Always check your answer !

$$P_{B' \leftarrow B} P_{B \leftarrow B'} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = I_3.$$

Thus the two matrices are inverses of one another:

$$P_{B' \leftarrow B} = (P_{B \leftarrow B'})^{-1}.$$

We now verify the change-of-basis formula.

Since the basis of the codomain has not changed,

$$[T]_{C \leftarrow B'} = [T]_{C \leftarrow B} P_{B \leftarrow B'}.$$

Indeed,

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 1 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & -1 \\ 3 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix},$$

which is exactly the matrix obtained in the previous question.

Therefore,

$$[T]_{C \leftarrow B'} = [T]_{C \leftarrow B} P_{B \leftarrow B'}.$$

7. Are the two matrix representations similar? Trick question :) No notion of matrix similarity applies here!

Recall that two square matrices $A, \tilde{A} \in \mathbb{R}^{n \times n}$ are similar if there exists an invertible matrix P such that

$$\tilde{A} = P^{-1}AP.$$

Similarity is therefore a notion for *square* matrices, and naturally arises when representing an endomorphism $T : V \rightarrow V$ in two different bases of the same vector space. Hence the two matrix representations are related by a change of basis, but not by a similarity transformation. $\square$

Exercise 2 (Rank of a matrix). Let

$$A = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix}.$$

1. Compute the rank of A using row reduction.
2. Find a basis for the column space of A .
3. Find a basis for the null space of A .
4. Determine if the columns of A are linearly independent.

Solution.

We are going to solve this exercise by first row-reducing A , then using this row-reduced form to answer the questions.

$$A = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix}.$$

First exchange R_1 and R_2 :

$$\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{pmatrix}.$$

Apply

$$R_3 \leftarrow R_3 + R_1 :$$

$$\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix}.$$

Then

$$R_3 \leftarrow R_3 - R_2 :$$

$$\begin{pmatrix} -1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

There are three pivots. Continuing to reduced row echelon form,

$$R_1 \leftarrow -R_1$$

gives

$$\begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$R_1 \leftarrow R_1 + 3R_3, \quad R_2 \leftarrow R_2 - 2R_3,$$

so

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

1. Since there are three pivot positions,

$$\boxed{\text{rank}(A) = 3}.$$

2. Every column is a pivot column.

Therefore the three original columns form a basis of the column space:

$$\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$$

is a basis of $\text{Col}(A)$.

Since the rank is 3,

$$\text{Col}(A) = \mathbb{R}^3.$$

3. Method 1: use the rank nullity theorem.

Method 2: Solve $Ax = 0$. Since the reduced row echelon form is I_3 , we obtain

$$x_1 = x_2 = x_3 = 0.$$

Thus

$$\ker(A) = \{0\}.$$

Its dimension is therefore

$$\dim(\ker A) = 0.$$

Important: The zero vector space has the empty family as a basis:

$$\text{a basis of } \ker(A) \text{ is } \emptyset.$$

□